

# पी एम् श्री केंद्रीय विद्यालय रानीखेत

Periodic Test-1

Class-11 Mathematics

TIME: 90 Min

M.M.40

**General Instructions:** All questions are compulsory.

| Q.No. | SECTION-A (1 Mark each)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1.    | Find the number of proper subsets of set of the set {a, b, c}.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1     |
| 2.    | If $\left(\frac{x}{3} + 1, y - \frac{2}{3}\right) = \left(\frac{5}{3}, \frac{1}{3}\right)$ , then find the values of x and y.                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1     |
| 3.    | <p>a. Assertion is correct, reason is correct; reason is a correct explanation for assertion.</p> <p>b. Assertion is correct, reason is correct; reason is not a correct explanation for assertion</p> <p>c. Assertion is correct, reason is incorrect</p> <p>d. Assertion is incorrect, reason is correct</p> <p><b>Assertion:</b> <math>\sec x</math> is negative in second and third quadrants.</p> <p><b>Reason:</b> <math>\cot x</math> decreases from 0 to <math>-\infty</math> in first quadrant and increases from 0 to <math>\infty</math> in third quadrant</p> | 1     |
| 4.    | The value of $(i^{15})^{-6} + \frac{1}{i^{-27}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |
| Q.No. | SECTION-B (2 Mark each)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
| 5.    | Let $A = \{1, 2, 3, 4, 5, 6\}$ , $B = \{2, 4, 6, 8\}$ . Find $A - B$ and $B - A$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |
| 6.    | Find the domain and range of the function $f(x) = \sqrt{36 - x^2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |
| 7.    | Find the angle in radian through which a pendulum swings if its length is 75 cm and tip describes an arc of length 21cm.                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2     |
| 8.    | If $(x + iy)(2 - 3i) = 4 + i$ , then find x and y.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |
| 9.    | Find the range of function $f(x) = \frac{ x-4 }{x-4}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     |
| Q.No. | SECTION-C (3 Mark each)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
| 10.   | Let A, B, C be the sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$ . Show that $B = C$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3     |
| 11.   | If $\tan x = \frac{3}{4}$ , $\pi < x < \frac{3\pi}{2}$ find the value of $\sin \frac{x}{2}$ , $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3     |
| 12.   | If $\alpha$ and $\beta$ are different complex numbers with $ \beta  = 1$ , then find $\left  \frac{\beta - \alpha}{1 - \bar{\alpha}\beta} \right $ .                                                                                                                                                                                                                                                                                                                                                                                                                      | 3     |
| 13.   | <p>Draw the graph of following function and write its domain and range.</p> $f(x) = \begin{cases} 1 - x, & \text{if } x < 0 \\ 1, & \text{if } x = 0 \\ x + 1, & \text{if } x > 0 \end{cases}$                                                                                                                                                                                                                                                                                                                                                                            | 3     |

| Q.No.             | SECTION-D (5 Mark each)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks                      |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 14.               | If $(x + iy)^3 = u + iv$ , then show that $\frac{u}{x} + \frac{v}{y} = 4(x^2 - y^2)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5                          |
| 15.               | a. Prove that $\sqrt{2 + \sqrt{2 + \sqrt{2 + 2 \cos 8A}}} = 2 \cos A$<br><b>OR</b><br>b. Prove that: $\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 60^\circ \cdot \sin 80^\circ = \frac{3}{16}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5                          |
| <b>CASE STUDY</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            |
| 16.               | <p>Trigonometry is the combination of 2 words – ‘Trigon’ means triangle and ‘metron’ means measure. It is a branch of geometry that studies relationship between lengths and angles of a triangle. Degree and radian units of measurement of angles are used, also called Indian system of measurement of triangles. In this system <math>\pi \text{ radian} = 180^\circ</math>; <math>1^\circ = 60 \text{ minute}</math> and <math>1 \text{ minute} = 60 \text{ seconds}</math>; <math>1 \text{ right angle} = 90^\circ</math>. The length of arc <math>l</math> is given by <math>l = r \theta</math>.</p> <p>On the basis of above information answer the following questions:</p> <p>(i) convert <math>\frac{11}{36}</math> radians into degree minutes and seconds.</p> <p>(ii) convert <math>\frac{7\pi}{18}</math> into degree measure</p> <p>(iii) Find the length of arc made by minute’s hand of a clock in 5 minutes having radius 7 cm.</p> <p><b>OR</b></p> <p>If the arcs of the same length in two circles subtend angles <math>65^\circ</math> and <math>80^\circ</math> at the centre, then the ratio of their radii.</p> | <p>1</p> <p>1</p> <p>2</p> |